Topic: Fourier sine series

Question: Find the Fourier sine series representation of $f(x) = x + x^3$ on $-L \leq x \leq L$.

Answer choices:

A $$f(x) = \frac{2L}{\pi^3} \sum_{n=1}^{\infty} \frac{(-1)^n}{n^3} (6L^2 - (n\pi L)^2 - (n\pi)^2) \cos\left(\frac{n\pi x}{L}\right) + \frac{L}{\pi}$$

B $$f(x) = \frac{2L}{\pi^3} \sum_{n=1}^{\infty} \frac{(-1)^n}{n^3} (6L^2 - (n\pi L)^2 - (n\pi)^2) \sin\left(\frac{n\pi x}{L}\right)$$

C $$f(x) = \frac{L}{\pi^2} \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} \sin\left(\frac{n\pi x}{L}\right)$$

D $$f(x) = 1 + \frac{L}{\pi^2} \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} \left(\sin\left(\frac{n\pi x}{L}\right) + n \cos\left(\frac{n\pi x}{L}\right)\right)$$

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Solution: B

Since $f(-x) = -f(x)$, the function is odd.

$$f(-x) = (-x) + (-x)^3$$

$$f(-x) = -x - x^3$$

$$-f(x) = -(x + x^3)$$

$$-f(x) = -x - x^3$$

For B_n, we get

$$B_n = \frac{2}{L}\int_0^L f(x)\sin\left(\frac{n\pi x}{L}\right)\,dx$$

$$B_n = \frac{2}{L}\int_0^L (x + x^3)\sin\left(\frac{n\pi x}{L}\right)\,dx$$

$$B_n = \frac{2}{L}\int_0^L x\sin\left(\frac{n\pi x}{L}\right)\,dx + \frac{2}{L}\int_0^L x^3\sin\left(\frac{n\pi x}{L}\right)\,dx$$

Use integration by parts with $u = x$, $du = dx$, $dv = \sin(n\pi x/L)\,dx$, and $v = -(L/n\pi)\cos(n\pi x/L)$.

$$B_n = -\frac{2Lx}{Ln\pi}\cos\left(\frac{n\pi x}{L}\right)\Big|_0^L + \frac{2L}{n\pi L}\int_0^L \cos\left(\frac{n\pi x}{L}\right)\,dx + \frac{2}{L}\int_0^L x^3\sin\left(\frac{n\pi x}{L}\right)\,dx$$

$$B_n = -\frac{2x}{n\pi}\cos\left(\frac{n\pi x}{L}\right) + \frac{2L}{(n\pi)^2}\sin\left(\frac{n\pi x}{L}\right)\Big|_0^L + \frac{2}{L}\int_0^L x^3\sin\left(\frac{n\pi x}{L}\right)\,dx$$

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$$B_n = -\frac{2L}{n\pi}\cos(n\pi) + \frac{2L}{(n\pi)^2}\sin(n\pi) - \left(0 + \frac{2L}{(n\pi)^2}\sin 0\right) + \frac{2}{L}\int_0^L x^3 \sin\left(\frac{n\pi x}{L}\right)\,dx$$

$$B_n = -\frac{2L}{n\pi}\cos(n\pi) + \frac{2}{L}\int_0^L x^3 \sin\left(\frac{n\pi x}{L}\right)\,dx$$

Now use integration by parts with $u = x^3$, $du = 3x^2\,dx$, $dv = \sin(n\pi x/L)\,dx$, and $v = -(L/n\pi)\cos(n\pi x/L)$.

$$\int_0^L x^3 \sin\left(\frac{n\pi x}{L}\right)\,dx = -\frac{Lx^3}{n\pi}\cos\left(\frac{n\pi x}{L}\right)\Big|_0^L + \frac{L}{n\pi}\int_0^L 3x^2\cos\left(\frac{n\pi x}{L}\right)\,dx$$

$$\int_0^L x^3 \sin\left(\frac{n\pi x}{L}\right)\,dx = -\frac{Lx^3}{n\pi}\cos\left(\frac{n\pi x}{L}\right)\Big|_0^L + \frac{3L}{n\pi}\int_0^L x^2\cos\left(\frac{n\pi x}{L}\right)\,dx$$

Now use integration by parts with $u = x^2$, $du = 2x\,dx$, $dv = \cos(n\pi x/L)\,dx$, and $v = (L/n\pi)\sin(n\pi x/L)$.

$$\int_0^L x^3 \sin\left(\frac{n\pi x}{L}\right)\,dx = \left(-\frac{Lx^3}{n\pi}\cos\left(\frac{n\pi x}{L}\right) + \frac{3L^2x^2}{(n\pi)^2}\sin\left(\frac{n\pi x}{L}\right)\right)\Big|_0^L$$

$$-\frac{6L^2}{(n\pi)^2}\int_0^L x\sin\left(\frac{n\pi x}{L}\right)\,dx$$

Now use integration by parts with $u = x$, $du = dx$, $dv = \sin(n\pi x/L)\,dx$, and $v = -(L/n\pi)\cos(n\pi x/L)$.

$$\int_0^L x^3 \sin\left(\frac{n\pi x}{L}\right)\,dx = -\frac{Lx^3}{n\pi}\cos\left(\frac{n\pi x}{L}\right) + \frac{3L^2x^2}{(n\pi)^2}\sin\left(\frac{n\pi x}{L}\right)$$

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$$+\frac{6L^3x}{(n\pi)^3}\cos\left(\frac{n\pi x}{L}\right)\Big|_0^L+\frac{6L^3}{(n\pi)^3}\int_0^L\cos\left(\frac{n\pi x}{L}\right)\,dx$$

$$\int_0^L x^3\sin\left(\frac{n\pi x}{L}\right)\,dx=-\frac{Lx^3}{n\pi}\cos\left(\frac{n\pi x}{L}\right)+\frac{3L^2x^2}{(n\pi)^2}\sin\left(\frac{n\pi x}{L}\right)$$

$$+\frac{6L^3x}{(n\pi)^3}\cos\left(\frac{n\pi x}{L}\right)+\frac{6L^4}{(n\pi)^4}\sin\left(\frac{n\pi x}{L}\right)\Big|_0^L$$

$$\int_0^L x^3\sin\left(\frac{n\pi x}{L}\right)\,dx=-\frac{L^4}{n\pi}\cos(n\pi)+\frac{3L^4}{(n\pi)^2}\sin(n\pi)+\frac{6L^4}{(n\pi)^3}\cos(n\pi)$$

$$+\frac{6L^4}{(n\pi)^4}\sin(n\pi)-\frac{6L^4}{(n\pi)^4}\sin(0)$$

$$\int_0^L x^3\sin\left(\frac{n\pi x}{L}\right)\,dx=-\frac{L^4}{n\pi}(-1)^n+\frac{6L^4}{(n\pi)^3}(-1)^n$$

Plugging this back into the equation for B_n, we get

$$B_n=-\frac{2L}{n\pi}\cos(n\pi)+\frac{2}{L}\left(-\frac{L^4}{n\pi}(-1)^n+\frac{6L^4}{(n\pi)^3}(-1)^n\right)$$

$$B_n=-\frac{2L}{n\pi}(-1)^n-\frac{2L^3}{n\pi}(-1)^n+\frac{12L^3}{(n\pi)^3}(-1)^n$$

$$B_n=\frac{2L}{(n\pi)^3}(-1)^n(6L^2-L^2(n\pi)^2-(n\pi)^2)$$

Then the Fourier sine series representation of $f(x)=x+x^3$ on $-L\le x\le L$ is

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$$f(x) = \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{L}\right)$$

$$f(x) = \frac{2L}{\pi^3} \sum_{n=1}^{\infty} \frac{(-1)^n}{n^3}(6L^2 - (n\pi L)^2 - (n\pi)^2)\sin\left(\frac{n\pi x}{L}\right)$$

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Topic: Fourier sine series

Question: Find the Fourier sine series representation of $f(x) = x - L$ on $0 \leq x \leq L$.

Answer choices:

A $$f(x) = -\frac{2L}{\pi}\sum_{n=1}^{\infty}\frac{1}{n}\sin\left(\frac{n\pi x}{L}\right)$$

B $$f(x) = \frac{L}{\pi^2}\sum_{n=1}^{\infty}\frac{(-1)^n - 1}{n^2}\sin\left(\frac{n\pi x}{L}\right)$$

C $$f(x) = \frac{L}{\pi^2} + \sum_{n=1}^{\infty}\frac{1}{n}\left(\sin\left(\frac{n\pi x}{L}\right) + \frac{(-1)^n}{n}\cos\left(\frac{n\pi x}{L}\right)\right)$$

D $$f(x) = 1 - \frac{\pi}{L} + \sum_{n=1}^{\infty}\frac{1}{n}\left(\sin\left(\frac{n\pi x}{L}\right) + \cos\left(\frac{n\pi x}{L}\right)\right)$$

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Solution: A

Since the function is not odd, let's find the function's odd extension on $-L \leq x \leq L$.

$$g(x) = \begin{cases} f(x) & 0 \leq x \leq L \\ -f(-x) & -L \leq x < 0 \end{cases}$$

$$g(x) = \begin{cases} x - L & 0 \leq x \leq L \\ x + L & -L \leq x < 0 \end{cases}$$

Because the odd extension is an odd function, we can find its Fourier sine series representation, starting with calculating B_n.

$$B_n = \frac{2}{L}\int_0^L f(x)\sin\left(\frac{n\pi x}{L}\right)\,dx$$

$$B_n = \frac{2}{L}\int_0^L (x - L)\sin\left(\frac{n\pi x}{L}\right)\,dx$$

$$B_n = \frac{2}{L}\int_0^L x\sin\left(\frac{n\pi x}{L}\right)\,dx - 2\int_0^L \sin\left(\frac{n\pi x}{L}\right)\,dx$$

Use integration by parts with $u = x$, $du = x\ dx$, $dv = \sin(n\pi x/L)\ dx$, and $v = -(L/n\pi)\cos(n\pi x/L)$.

$$B_n = \frac{2}{L}\left[-\frac{Lx}{n\pi}\cos\left(\frac{n\pi x}{L}\right) + \int \frac{L}{n\pi}\cos\left(\frac{n\pi x}{L}\right)\,dx\right]\Bigg|_0^L + \frac{2L}{n\pi}\cos\left(\frac{n\pi x}{L}\right)\Bigg|_0^L$$

$$B_n = \frac{2}{L}\left[-\frac{Lx}{n\pi}\cos\left(\frac{n\pi x}{L}\right) + \frac{L^2}{(n\pi)^2}\sin\left(\frac{n\pi x}{L}\right)\right]\Bigg|_0^L + \frac{2L}{n\pi}\cos\left(\frac{n\pi x}{L}\right)\Bigg|_0^L$$

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$$B_n = -\frac{2x}{n\pi}\cos\left(\frac{n\pi x}{L}\right) + \frac{2L}{(n\pi)^2}\sin\left(\frac{n\pi x}{L}\right) + \frac{2L}{n\pi}\cos\left(\frac{n\pi x}{L}\right)\Bigg|_0^L$$

$$B_n = \frac{2L-2x}{n\pi}\cos\left(\frac{n\pi x}{L}\right) + \frac{2L}{(n\pi)^2}\sin\left(\frac{n\pi x}{L}\right)\Bigg|_0^L$$

Evaluate over the interval.

$$B_n = \frac{2L-2L}{n\pi}\cos\left(\frac{n\pi L}{L}\right) + \frac{2L}{(n\pi)^2}\sin\left(\frac{n\pi L}{L}\right)$$

$$-\left[\frac{2L-0}{n\pi}\cos\left(\frac{n\pi(0)}{L}\right) + \frac{2L}{(n\pi)^2}\sin\left(\frac{n\pi(0)}{L}\right)\right]$$

$$B_n = \frac{2L}{(n\pi)^2}\sin(n\pi) - \frac{2L}{n\pi}$$

$$B_n = -\frac{2L}{n\pi}$$

Then the Fourier sine series representation of $f(x) = x - L$ on $0 \le x \le L$ is

$$f(x) = \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{L}\right)$$

$$f(x) = -\frac{2L}{\pi}\sum_{n=1}^{\infty}\frac{1}{n}\sin\left(\frac{n\pi x}{L}\right)$$

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Topic: Fourier sine series

Question: Find the Fourier sine series representation of $f(x) = e^x$ on $0 \leq x \leq L$.

Answer choices:

A $$f(x) = \pi + \sum_{n=1}^{\infty} \frac{n}{(n\pi)^2 + L} \left((-1)^n \cos\left(\frac{n\pi x}{L}\right) + \sin\left(\frac{n\pi x}{L}\right) \right)$$

B $$f(x) = \frac{\pi}{L^2} \sum_{n=1}^{\infty} n(-1)^n (e^L - 1) \sin\left(\frac{n\pi x}{L}\right)$$

C $$f(x) = \sum_{n=1}^{\infty} \frac{2n\pi(e^L(-1)^{n+1} + 1)}{(n\pi)^2 + L} \cos\left(\frac{n\pi x}{L}\right)$$

D $$f(x) = \sum_{n=1}^{\infty} \frac{2n\pi(e^L(-1)^{n+1} + 1)}{(n\pi)^2 + L^2} \sin\left(\frac{n\pi x}{L}\right)$$

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Solution: D

Since the function is not odd, let's find the function's odd extension on $-L \leq x \leq L$.

$$g(x) = \begin{cases} f(x) & 0 \leq x \leq L \\ -f(-x) & -L \leq x < 0 \end{cases}$$

$$g(x) = \begin{cases} e^x & 0 \leq x \leq L \\ -e^{-x} & -L \leq x < 0 \end{cases}$$

Because the odd extension is an odd function, we can find its Fourier sine series representation, starting with calculating B_n.

$$B_n = \frac{2}{L}\int_0^L f(x)\sin\left(\frac{n\pi x}{L}\right)\,dx$$

$$B_n = \frac{2}{L}\int_0^L e^x \sin\left(\frac{n\pi x}{L}\right)\,dx$$

Use integration by parts with $u = e^x$, $du = e^x\,dx$, $dv = \sin(n\pi x/L)\,dx$, and $v = -(L/n\pi)\cos(n\pi x/L)$.

$$B_n = \frac{2}{L}\left[-\frac{Le^x}{n\pi}\cos\left(\frac{n\pi x}{L}\right) + \frac{L}{n\pi}\int e^x \cos\left(\frac{n\pi x}{L}\right)\,dx\right]\Bigg|_0^L$$

$$B_n = \left[-\frac{2e^x}{n\pi}\cos\left(\frac{n\pi x}{L}\right) + \frac{2}{n\pi}\int e^x \cos\left(\frac{n\pi x}{L}\right)\,dx\right]\Bigg|_0^L$$

Use integration by parts with $u = e^x$, $du = e^x\,dx$, $dv = \cos(n\pi x/L)\,dx$, and $v = (L/n\pi)\sin(n\pi x/L)$.

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$$B_n = \left[-\frac{2e^x}{n\pi} \cos\left(\frac{n\pi x}{L}\right) + \frac{2Le^x}{(n\pi)^2} \sin\left(\frac{n\pi x}{L}\right) - \frac{2L}{(n\pi)^2} \int e^x \sin\left(\frac{n\pi x}{L}\right) \, dx \right] \Bigg|_0^L$$

$$B_n = \left[-\frac{2e^x}{n\pi} \cos\left(\frac{n\pi x}{L}\right) + \frac{2Le^x}{(n\pi)^2} \sin\left(\frac{n\pi x}{L}\right) \right] \Bigg|_0^L - \frac{L^2}{(n\pi)^2} B_n$$

$$B_n \left(1 + \frac{L^2}{(n\pi)^2}\right) = \left[-\frac{2e^x}{n\pi} \cos\left(\frac{n\pi x}{L}\right) + \frac{2Le^x}{(n\pi)^2} \sin\left(\frac{n\pi x}{L}\right) \right] \Bigg|_0^L$$

Evaluate over the interval.

$$B_n \left(1 + \frac{L^2}{(n\pi)^2}\right) = -\frac{2e^L}{n\pi} \cos(n\pi) + \frac{2Le^L}{(n\pi)^2} \sin(n\pi) + \frac{2}{n\pi}$$

$$B_n \left(1 + \frac{L^2}{(n\pi)^2}\right) = \frac{2e^L(-1)^{n+1} + 2}{n\pi}$$

$$B_n = \frac{2e^L(-1)^{n+1} + 2}{n\pi} \left(\frac{(n\pi)^2}{(n\pi)^2 + L^2} \right)$$

$$B_n = \frac{2n\pi(e^L(-1)^{n+1} + 1)}{(n\pi)^2 + L^2}$$

Then the Fourier sine series representation of $f(x) = e^x$ on $0 \le x \le L$ is

$$f(x) = \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{L}\right)$$

$$f(x) = \sum_{n=1}^{\infty} \frac{2n\pi(e^L(-1)^{n+1} + 1)}{(n\pi)^2 + L^2} \sin\left(\frac{n\pi x}{L}\right)$$

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